PostgreSQL

Delivered

DATABASE

Below are the 04 main advantages.

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TODAY

Deliver the Reliable Database.

Let's finish verification and give 04 main advantages

1/ What is the extent of missing data in the other time series?

	series_id text	n_jumps bigint
1	UMCSENTX	74
2	CE16OV	1
3	CLF16OV	1
4	CPIAPPSL	1
5	CPIAUCSL	1
6	CPIMEDSL	1
7	CPITRNSL	1
8	CPIULFSL	1
9	CUSR0000SA0...	1
10	CUSR0000SA0...	1
11	CUSR0000SAC	1
12	CUSR0000SAD	1
13	CUSR0000SAS	1
14	M2REAL	1

All variables exhibit a single jump, in October 2025, except UMCSENTX

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2/ Where does the gap occur for UMCSENTX and others ?

	series_id [PK] text	prev_date date	date [PK] date	month_diff integer
71	UMCSENTX	1977-02-01	1977-05-01	3
72	UMCSENTX	1977-05-01	1977-08-01	3
73	UMCSENTX	1977-08-01	1977-11-01	3
74	UMCSENTX	1977-11-01	1978-01-01	2
75	CE16OV	2025-09-01	2025-11-01	2
76	CLF16OV	2025-09-01	2025-11-01	2
77	CPIAPPSL	2025-09-01	2025-11-01	2
78	CPIAUCSL	2025-09-01	2025-11-01	2
79	CPIMEDSL	2025-09-01	2025-11-01	2
80	CPITRNSL	2025-09-01	2025-11-01	2
81	CPIULFSL	2025-09-01	2025-11-01	2
82	CUSR0000SA0...	2025-09-01	2025-11-01	2
83	CUSR0000SA0...	2025-09-01	2025-11-01	2
84	CUSR0000SAC	2025-09-01	2025-11-01	2

Structural jump in October 2025 across all variables, except UMCSENTX.

3/ UMCSENTX: is this a gap rather than a jump ?

It measures U.S. consumer sentiment on economic / financial conditions

No problem. it Collected quarterly by the University of Michigan before 1978.

4/ How many variables are missing in October 2025 ?

Query Query History

```sql
SELECT COUNT(*) AS n_missing_series
FROM macro.series AS s
LEFT JOIN macro.observations_monthly AS o
  ON o.series_id = s.series_id
 AND o.date = DATE '2025-10-01'
WHERE o.series_id IS NULL
   OR o.value IS NULL;
```

Data Output Messages Notifications

SQL

Showing rows: 1 to 1 Page No: 1 of 1

	n_missing_series bigint
1	49

49/126 variables miss in October 2025

-So data are stopped in September 2025

5/ What are the impact of this stop on the AI system ?

Build production rapidly and reliably:

- *Experimentation*
- *Staging*
- *Production*

Production requirements

- *Monitor and trigger data freshness via the FRED API*
- *If updated (eg. Oct 2025) → data ingestion*
- *Else → time-series–aware imputation → data ingestion*

6/ Are all observations aligned to monthly dates ?

Query Query History

```sql
SELECT
    COUNT(*) FILTER (
        WHERE date = DATE_TRUNC('month', date)
    ) AS aligned,
    COUNT(*) AS total
FROM macro.observations_monthly;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	aligned bigint	total bigint
1	101981	101981

Dates are at the beginning of the month.

6/ How many missing values are there in total ?

Query   Query History

```sql
SELECT
    COUNT(*) AS total_rows,
    COUNT(*) FILTER (WHERE value IS NULL) AS n_missing,
    ROUND(
        100.0 * COUNT(*) FILTER (WHERE value IS NULL) / COUNT(*),
        2
    ) AS missing_rate_pct
FROM macro.observations_monthly;
```

Data Output   Messages   Notifications

SQL

	total_rows bigint	n_missing bigint	missing_rate_pct numeric
1	101981	1029	1.01

1.01% of observations are missing
1029 missing observations

7/ Which are the top 10 least complete variables ?

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	series_id text	total_obs bigint	n_nan bigint	pct_nan numeric
1	ACOGNO	803	399	49.69
2	TWEXAFEGSMT...	803	168	20.92
3	UMCSENTX	803	154	19.18
4	ANDENOX	803	110	13.70
5	VIXCLSX	803	42	5.23
6	PERMITMW	803	15	1.87
7	PERMITNE	803	15	1.87
8	PERMIT	803	15	1.87
9	PERMITS	803	15	1.87
10	PERMITW	803	15	1.87

8/ Number of NaNs in commonly used variables in the literature?

SQL

Showing rows: 1 to 10 Page No: 1 of 1

	series_id text	total_obs bigint	n_nan bigint	pct_nan numeric
1	BUSLOANS	801	0	0.00
2	CPIAUCSL	801	0	0.00
3	DPCERA3M086SB...	801	0	0.00
4	INDPRO	801	0	0.00
5	M2SL	801	0	0.00
6	OILPRICEX	801	0	0.00
7	RPI	801	0	0.00
8	SP500	801	0	0.00
9	TB3MS	801	0	0.00
10	UNRATE	801	0	0.00

Most usefull Variables doesn't have missing values

Agnostic Time Series

Targets and features can be interchanged without changing the pipeline.

01

Series_ID	Date	**Value**
....	...	...

Scalability and reusability

New series and variables are added without redesign.

02

Series_ID	Date	**Value**
....	...	...

Pannel-Ready

Supports multi-entity and cross-country economic analysis.

03

Entity	Series_ID	Date	**Value**
....	...	...	...

Multi-usage

04